Multi-Stage Financial Intelligence System: Extractive QnA, Summarization, and Agentic Analysis

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***Abstract*—The interpretation of high-density financial data is a critical bottleneck in professional market analysis. This project presents a multi-layered NLP system designed to automate extraction, summarization, and trend inference from 10-K filings. We evaluate an extractive QnA module using the FinanceBench dataset and a summarization engine using the ECTSum dataset.**

***Index Terms*—NLP, Financial Analysis, RAG, RAGAs, G-Eval, BERTScore, Summarization, FinanceBench.**

I. Introduction

The volume of financial documentation, such as SEC 10-K and 10-Q filings, poses a significant challenge for human analysts. Traditional information retrieval often misses the semantic nuances and the structural relationships within financial tables. This project develops a modular pipeline that progresses from simple data extraction (Module 1) to summarization (Module 2) and ultimately to multi-document agentic reasoning (Module 3).

II. Module 1: Extractive Question Answering

The first module is designed to provide precise, evidence-backed answers to specific financial questions using a retrieval-augmented approach.

A. Dataset: FinanceBench

We utilized the **FinanceBench** dataset, a rigorous benchmark containing 150 samples of questions, ground truth answers, and the specific document evidence required. The dataset is particularly effective because it tests a model's ability to distinguish between distinct but related metrics, such as *Operating Income* and *Comprehensive Income*.

B. Exploratory Data Analysis (EDA)

Our EDA of the FinanceBench dataset revealed that answers are not uniformly "short."

- **Answer Length Distribution:** A significant portion of the answers (approx. 75%) are long-form reasoning sentences, while only 25% are short numerical extractions (≤ 5 words).
- **Document Context:** The average evidence length is over 500 words, requiring robust context-handling capabilities.

TABLE I
Module 1: FinanceBench EDA Summary

Metric	Value
Total Samples	150
Avg. Context Length	501 words
Max Context Length	1,564 words
Answer Type: Short (Extraction)	38
Answer Type: Long (Reasoning)	112

C. Architecture

The module utilizes a **RAG (Retrieval-Augmented Generation)** framework. Documents were vectorized using Google *text-embedding-004* into a ChromaDB instance. We utilized **Gemini 2.0 Flash** as the reasoning engine.

We used RAGAs as our evaluation method. RAGAs (Retrieval-Augmented Generation Assessment) is an open-source framework designed to evaluate RAG pipelines by measuring the quality of both the retrieved context and the generated response. It utilizes large language models (LLMs) as "judges" to provide objective scores on metrics such as faithfulness, answer relevance, and context precision, reducing the need for expensive manual human annotation.

Initial zero-shot attempts resulted in insufficient RAGAs scores, particularly regarding faithfulness. To resolve this, we implemented various prompting techniques such as **Role**, **Chain-of-Thought (CoT)** and **Few-Shot prompting**. This approach forces the model to articulate the reasoning steps and locate specific line items in the text before formulating the final answer.

The specific prompt structure utilized to implement this Chain-of-Thought and Few-Shot strategy is detailed below:

```
cot_few_shot_prompt = """
You are an expert financial analyst.
Answer the question based strictly on the
    context.
```

```
Guidelines:
1. Think step-by-step before answering.
2. If it's a table, identify the correct row
    and column.
3. Finally, provide the concise answer at the
    end.

---
Example 1 (Table Extraction):
Context: "Revenue (Millions) 2020 | 2021 $10.5
     | $12.0"
Question: What was the 2021 revenue?
Reasoning: The table shows columns for 2020
    and 2021. The Revenue row intersects with
    2021 at $12.0.
Answer: $12.0 million

Example 2 (Text Reasoning):
Context: "Operating margin declined 200 bps
    primarily due to higher raw material costs
    , specifically aluminum."
Question: What drove the decline in operating
    margin?
Reasoning: The text explicitly states the
    decline was "primarily due to higher raw
    material costs."
Answer: Higher raw material costs,
    specifically aluminum.

Example 3 (Calculation/Comparison):
Context: "Net Income was $5M in 2020 and $7M
    in 2021."
Question: Did Net Income increase?
Reasoning: Comparing 2021 ($7M) to 2020 ($5M),
     the value is higher.
Answer: Yes, Net Income increased from $5M to
    $7M.
---

Context: {context}
Question: {question}
Reasoning:"""
```

D. Evaluation Results

The evaluation leveraged RAGAs (Faithfulness, Answer Relevancy, Context Precision), BERTScore, and Exact Match (EM).

TABLE II
MODULE 1: EXTRACTIVE QNA EVALUATION METRICS

Metric	Initial (Zero-Shot)	Final (CoT + FS)
RAGAs Faithfulness	0.1800	0.7393
RAGAs Answer Relevancy	0.0000	0.4476
RAGAs Context Precision	0.0000	0.7667
BERTScore F1	0.8000	0.8800
Exact Match	0.0000	0.1100

The **Exact Match** score remained low at 0.11. This is attributed to the rigidity of the metric; financial data can be formatted in multiple valid ways (e.g., "$10.5M" vs. "10,500,000"). EM treats any character-level variance as a failure. However, the **BERTScore F1** of 0.88 suggests high semantic alignment.

The RAGAs Faithfulness (0.7393) and Context Precision (0.7667) demonstrate that the transition to Few-Shot and CoT prompting successfully grounded the model in the retrieved evidence, significantly reducing hallucinations. The Answer Relevancy (0.4476) is lower because the model often includes additional context to ensure clarity, whereas ground truth answers in FinanceBench are often more terse.

III. MODULE 2: FINANCIAL SUMMARIZATION

The second module addresses the need for comprehensive summaries of lengthy earnings transcripts and reports.

A. Dataset: ECTSum

We used the test set of the **ECTSum** dataset, consisting of 495 instances. A major challenge identified in this dataset is the size of the ground truth summaries. On average, transcripts contain 2,800 words, while ground truth summaries average only 44 words. This extreme compression ratio (60:1) highlights a significant divergence in utility; while traditional metrics penalize models for deviating from the brevity of the ground truth, longer summaries often provide greater analytical value. In a professional financial context, very short summaries risk omitting the nuanced operational drivers and specific quantitative guidance required for informed decision-making. By maintaining a more comprehensive summary length, the model is able to capture the "why" behind the numbers, such as specific headwinds in supply chain logistics or regional margin fluctuations, which are typically lost in high-compression references.

B. Architecture

We employed the **Gemini 2.0 Flash** model with a structured prompt template.

The summarization module also utilizes a structured prompt designed to transition the LLM into a "Senior Financial Analyst" persona. The use of essentail Prompt Engineering techniques such as role prompting and detailed unambiguous instructions enhanced the output of the system. To ensure consistency across 495 transcripts, the following template was used:

```
summary_prompt = """
You are a Senior Financial Analyst.
Your task is to summarize the following
    Earnings Call Transcript into a
    professional investment memo.

Guidelines:
1. Financial Performance: Extract key numbers
    (Revenue, EPS, Margins) and compare with
    previous periods if mentioned.
2. Future Guidance: Summarize the CEO/CFO's
    outlook for the next quarter/year.
3. Key Risks/Highlights: Mention any major
    operational changes, headwinds, or
    tailwinds.
4. Tone: Professional, objective, and dense
    with facts.
```

```
5. Length: Approximately 200-300 words. Do not
    be overly brief.

Transcript:
{text}

Investment Memo:
"""
```

C. Evaluation Metrics

Evaluation was conducted using **ROUGE (1, 2, L)**, **BERTScore**, and **G-Eval**.

- **ROUGE Discrepancy:** ROUGE scores were significantly low (0.08 for ROUGE-L). This is attributed to the dataset's high compression; the LLM included critical financial details (like specific percentage savings in warehouse management) that the human-written summaries omitted for extreme brevity.
- **BERTScore:** BERTScore remained high (> 0.8), indicating that the semantic meaning was highly aligned despite different word choices.
- **Reliance on G-Eval:** Due to the limitations of ROUGE, we relied on **G-Eval** (LLM-as-a-judge). G-Eval confirmed that the model's summaries were coherent (4.4/5.0) and contextually accurate.

TABLE III
MODULE 2: SUMMARIZATION EVALUATION

Metric	Value
ROUGE-L	0.08
BERTScore	0.82
G-Eval	4.39 / 5.0

IV. MODULE 3: AGENTIC FINANCIAL ANALYZER

The final stage of the pipeline addresses the "Inference Gap", i.e. the inability of standard RAG systems to perform cross-document reasoning or high-precision mathematical calculations. We developed an autonomous financial research agent capable of multi-step planning and tool-augmented reasoning to provide deep analysis beyond simple fact extraction.

A. Dataset: 50 Multi-Year SEC Filings

To evaluate the system's ability to track trends over time, we curated a database of 50 financial reports. This consists of the annual 10-K filings for 10 leading companies (including Apple, Microsoft, Tesla, and Nvidia) spanning a 5-year period.

The scale of these documents presents a significant "needle in a haystack" challenge. On average, a single SEC 10-K filing contains between 40,000 to 50,000 words, translating to approximately 120,000 to 160,000 tokens depending on the complexity of the footnotes and tables. Processing 50 such documents requires a system that can effectively filter noise and target specific semantic clusters across millions of tokens.

B. Ingestion Methodology: Semantic Element Partitioning

A major problem faced during ingestion was the structural degradation of financial tables in SEC filings. Standard recursive character splitting often breaks tables mid-row, rendering them unreadable by LLMs.

- **Direct HTML Parsing:** We utilized `BeautifulSoup` to isolate the `<DOCUMENT>` and `<TEXT>` tags specifically within 10-K filings, discarding irrelevant metadata and header noise.
- **Markdown Table Conversion:** To maintain the spatial relationship between financial metrics and fiscal years, we implemented a custom `html_to_markdown_table` function. This converts HTML `<tr>` and `<td>` tags into Markdown strings (|Header|Value|).
- **Vector Storage:** Chunks were embedded using `text-embedding-004` and stored in a Chroma DB with high-fidelity metadata (Ticker and Year), allowing for strict filtering during the retrieval phase.

C. Architecture: Plan-and-Execute

Rather than acting as a simple conversational chatbot that answers immediately, the analyzer functions as a systematic researcher. It follows a structured logic loop that forces the system to first map out the necessary research steps, such as identifying which tables to read and which calculations to perform, before attempting to generate a final response.

*1) **The Planner and System Prompt**:* Upon receiving a query, the `Gemini-2.5-flash-lite` model acts as a "Lead Researcher." It does not answer immediately; instead, it drafts a step-by-step execution plan. The following system prompt was engineered to enforce this rigorous analytical behavior:

```
system_prompt = """
You are a Lead Financial Research Agent. You
    solve complex queries by creating a '
    Research Plan'.

STRICT OPERATING PROCEDURE:
1. Analyze the Goal: Determine if the user
    wants a single data point, a calculation,
    or a multi-year comparison.
2. Draft a Plan:
   - Step 1: Identify and retrieve the
      specific 10-K tables needed for all
      years involved.
   - Step 2: Use the python_calculator for
      math.
   - Step 3: Search for qualitative sections (
      MD&A, Risk Factors) to find the 'Why'.
3. Execute: Run the steps sequentially.
4. Self-Correct: If a table looks incomplete,
    try a more specific search query.

RATIO FORMULAS:
- Acid Test: (Total Current Assets -
    Inventories) / Total Current Liabilities.
- Current Ratio: Total Current Assets / Total
    Current Liabilities.
```

```
- Net Margin: Net Income / Total Net Sales.

Always explain the formula you used and list
    the raw numbers found before showing the
    final result.
"""
```

*2) **Tool Augmentation**:* The agent is equipped with two specialized tools:

- **search_10k_reports:** A structured tool that uses similarity search with a metadata filter. It is enhanced to automatically append keywords like "Management's Discussion and Analysis" when qualitative reasoning is required.
- **python_calculator:** A `PythonREPL` tool that executes safe code for financial math, ensuring that ratios are calculated with floating-point precision rather than LLM guesswork.

D. Conversational Clarification and Chatbot Logic

A critical feature implemented in Module 3 is the **Clarification Loop**. In a real-world scenario, users often provide vague queries such as "What was the net revenue?" without specifying a company or a fiscal year.

Since the analyzer relies on strict metadata filtering (`Ticker` and `Year`) to ensure accuracy and prevent cross-contamination of data, the agent is programmed to recognize missing parameters. Instead of hallucinating a generic answer or searching the entire database, the chatbot functionality engages the user to provide the missing metadata. This validation step ensures the system only proceeds when it has the precise metadata required to target specific evidence. This validation step ensures the system only proceeds when it has the exact metadata required to isolate the correct evidence. By enforcing these parameters, the system eliminates the risk of cross-context contamination, where data from different companies or periods might be incorrectly merged. This rigorous factual standard ensures the analyzer only operates when the query context is fully defined, preventing errors by requiring specific parameters rather than relying on estimation.

V. Evaluation

To assess the reliability of the Module 3 pipeline, we developed a custom benchmarking mini dataset designed to test multi-step reasoning, mathematical precision, and tool-use accuracy.

A. Evaluation Dataset Methodology

The evaluation dataset was constructed to simulate the complex, multi-layered queries typical of equity research.

- **Creation Process:** We scanned the 50 10-K filings in our database and identified 20 target metrics. For each metric, a ground truth was established by manually examining the original SEC filings. This included identifying the exact table, row, and footnote required for a correct answer.
- **Composition:** The set of 20 questions was stratified into three categories:
  - **Pure Calculation:** Requires identifying two or more metrics and applying a formula.
  - **Pure Textual Reasoning:** Requires synthesizing qualitative commentary from the MD&A.
  - **Mixed:** Requires calculating a ratio and then explaining the operational "Why" behind the number.

B. Performance Metrics: Functional Call Accuracy (FCA)

In addition to answer correctness, we introduced **Functional Call Accuracy (FCA)**. This metric measures the agent's decision-making integrity.

- **Definition:** FCA is the ratio of queries where the agent successfully invoked the required set of tools (e.g., calling both the search tool and the calculator for a "Mixed" query) vs. the total queries.
- **Calculation:** $FCA = \frac{\text{Successful Tool Paths}}{\text{Total Queries}} \times 100$.
- **Results:** The system achieved an **FCA of 60.00%**. This indicates that while the agent is highly capable, it occasionally defaults to conversational "chatbot" behavior, opting to ask for clarification rather than utilizing the data provided in the prompt.

C. Manual Evaluation and Error Analysis

Table IV provides a manual comparison between the ground truth and the system-generated responses.

D. Discussion of Accuracy

The final accuracy of 60.00% represents a conservative lower bound. In cases of disagreement, the failure was rarely a hallucination (false data). Instead, the system primarily failed due to **Conversational Guardrails** where the agent asked for user confirmation for tickers already present in the query. This demonstrates that the system is safe but requires further prompt tuning to increase its autonomy.

VI. Conclusion and Future Work

This project successfully developed a multi-stage financial intelligence framework that bridges the gap between raw, unstructured corporate filings and actionable investment insights. By modularizing the pipeline into extractive QnA, executive summarization, and autonomous agentic analysis, we addressed the specific challenges of data density and numerical precision inherent in financial NLP.

A. Key Findings

Our investigation yielded three primary conclusions regarding the deployment of Large Language Models (LLMs) in the financial domain:

- **Limitations of Traditional Metrics:** As established in Modules 1 and 2, traditional metrics such as Exact Match (EM) and ROUGE are insufficient for financial contexts. EM fails to account for varied numerical formatting, while ROUGE penalizes models for generating comprehensive summaries that exceed the brevity of highly compressed ground-truth datasets.

TABLE IV
MODULE 3: COMPREHENSIVE ANALYZER BENCHMARK AND ERROR ANALYSIS

Question Topic	Type	Agreement	Explanation for Disagreement
AAPL (2023) Current Ratio	Pure Calculation	**Yes**	N/A (Perfect Numerical Match).
AMZN (2022) Net Profit Margin	Pure Calculation	**Yes**	N/A (Correct formula and sign retention).
TSLA (2023) Asset Growth	Pure Calculation	**Yes**	N/A (Multi-year comparison correct).
MSFT (2023) Debt-to-Equity	Pure Calculation	**Yes**	N/A (Balance sheet extraction correct).
NVDA (2024) Inv. Turnover	Pure Calculation	**Yes**	N/A (Multi-step logic preserved).
META (2023) Competition	Reasoning	**No**	**Chatbot Logic:** Agent requested ticker instead of searching.
GOOGL (2022) AI Strategy	Reasoning	**Yes**	N/A (Successfully synthesized MD&A).
NFLX (2023) Content Spend	Reasoning	**Yes**	N/A (Correctly identified cash flow drivers).
V (2023) Cross-border Recovery	Reasoning	**Yes**	N/A (Numerical-text synthesis match).
JPM (2023) Cyber Risks	Reasoning	**Yes**	N/A (Extracted specific risk clusters).
AAPL (2023) R&D % of Sales	Mixed	**Yes**	N/A (Correct formula and driver extraction).
AMZN (2023) ROA & Efficiency	Mixed	**No**	**Chatbot Logic:** Agent asked for ticker symbol.
TSLA (2022) Gross Margin	Mixed	**No**	**Missing Data:** 2022 10-K was not indexed in DB.
NVDA (2024) Cash Position	Mixed	**No**	**Incomplete Chain:** Agent drafted plan but didn't finish calculation.
META (2023) Op. Margin	Mixed	**Yes**	N/A (Correctly linked calculation to efficiency gains).
MSFT (2023) Cloud Growth	Mixed	**No**	**Entity Mismatch:** Agent used "Intelligent Cloud" instead of "Microsoft Cloud".
NFLX (2022) FCF vs Net Inc	Mixed	**No**	**Chatbot Logic:** Agent asked for a stock ticker.
GOOGL (2023) Tax Rate	Mixed	**Yes**	N/A (Correct plan and numerical reasoning).
V (2022) Dividend Payout	Mixed	**No**	**Chatbot Logic:** Agent asked for missing parameters.
JPM (2022) CET1 Ratio	Mixed	**No**	**Chatbot Logic:** Agent asked bank ticker/name.

- **Efficacy of Semantic Evaluation:** The transition to model-based evaluation frameworks, specifically **RAGAs** and **G-Eval**, provided a more nuanced understanding of system performance. These metrics allowed us to measure *Faithfulness* and *Coherence*, ensuring that the intelligence generated was not only semantically aligned but factually grounded in the source documents.
- **Autonomy vs. Guardrails:** The development of the Agentic Analyzer (Module 3) demonstrated that a **Plan-and-Execute** architecture, supported by **Semantic Element Partitioning**, is capable of high-precision math. However, the benchmark results revealed a tension between conversational safety guardrails and autonomous execution, which is a design-specific choice limited to only our implementation. Our observed **Functional Call Accuracy (FCA) of 60.00%** was largely influenced by the agent's cautious clarification logic rather than factual errors.

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